

Time evolutions of scalar field perturbation in Schwarzschild de-Sitter black hole from Einstein-scalar–Gauss–Bonnet theory

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Abstract

The scalar-free black hole could be unstable against the scalar field perturbation when it is coupled to a Gauss–Bonnet (GB) invariant in a special form. It is known that the tachyonic instability in this scenario is triggered by the sufficiently strong GB coupling. In this paper, we focus on the time domain analysis of massive scalar field perturbation around the Schwarzschild de-Sitter black hole in Einstein-scalar–Gauss–Bonnet gravity. By analyzing the scalar field propagation, we find that the scalar field will finally grow when the GB coupling is large enough. This could lead to the instability of the background black hole. Furthermore, we demonstrate how the mass of the scalar field and the GB coupling strength affect the onset of tachyonic instability.

Keywords: black hole perturbation, scalar-Einstein-Gauss-Bonnet gravity, time domain evolution

(Some figures may appear in colour only in the online journal)

1. Introduction

A new era of exploring the strong gravity regime is coming along with the detection of gravitational waves [1, 2] and black hole images [3]. In the theoretical aspect, higher-order curvature terms are significant for the study of the strong gravity regime. Such well-known terms are the Gauss–Bonnet (GB) correction, which is always coupled with the scalar field to hold their contribution to the dynamic of a four-dimensional spacetime [4]. Recently, a specific Einstein-scalar-Gauss–Bonnet (EsGB) theory, in which a special formula coupling between the scalar field and the GB term is added into the ordinary Einstein–Hilbert action, has recently gained a lot of attention since it evades the famous no-hair theorem. In this theory, a hairy solution could develop by the so-called spontaneous scalarization process [5–7]. In particular, below a certain mass, the Schwarzschild black hole background may

become unstable in regions of strong curvature, resulting in the formation of a scalarized hairy black hole when the scalar field backreacts to the geometric. This proposal on spontaneous scalarization has inspired numerous generalizations, see [8–35] for examples and references therein.

In addition, because de-Sitter spacetime has cosmological advantages [36, 37] and de-Sitter/conformal field theory implications [38, 39], spontaneous scalarization in EsGB theory has been extensively studied in dS spacetime [40, 41]. The authors of [40] proposed that a regular black hole horizon with non-trivial hair could form. However, there was no complete hairy solution and the underlying reason remains unknown. Later on, it was stated in [41] that if the scalar field is confined between two horizons of a dS black hole, then a hairy black hole is unlikely to form; if the scalar field is allowed to extend beyond the cosmological horizon, a new hairy black hole was numerically constructed. Those findings suggest that the existence of a cosmological horizon, in fact, results in new points in the scalarization of black holes.

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The analysis of (in)stability of the scalar field on the background was present in [40, 41], but they only considered the massless and r -dependent scalar field. It is natural to include a mass term or a more general self-interaction term for the scalar field from the viewpoint of field theory. Furthermore, it has been discovered that the scalar field's mass suppresses the scalar radiation with gravitational waves [42, 43]. The effect of mass on spontaneous scalarization in EsGB theory in asymptotical flat spacetime has also been studied, please refer to [44–46] for instance. All of this encourages us to investigate the influence of the mass term on scalar field instability in the dS spacetime. Moreover, it is also important to investigate the dynamical evolution of a time and space-dependent scalar field. According to [47], the time evolution of the perturbations in the black hole background generally goes through three stages. Following the initial burst, the perturbation experiences damped oscillations, which are characterized by the quasinormal ringing with frequencies determined by the black hole parameters. At later times, quasinormal oscillations are swamped by the relaxation process, which is the requirement of the black hole no-hair theorem.

Thus, we will consider a massive scalar field as a probe field on the Schwarzschild de-Sitter (SdS) black hole in this paper. We will study the scalar field dynamics by solving the linear equation of motion in the time domain, which will give us a picture of how the field propagates to grow up and eventually trigger the tachyonic instability. We will see that the GB coupling and the mass of the scalar field affect the onset of the tachyonic instability, which could help us understand the scalarization of the dS black hole in the special EsGB theory.

The remainder of this paper is organized as follows. In section 2, we show the covariant equation for a probed massive scalar field in a special EsGB gravity. Then we give a preliminary instruction on the study of wave dynamics. In section 3, we show the results of the scalar field propagation and analyze the onset of tachyonic instability of the SdS black hole. The last section is our closing remarks. We shall work with the units $\hbar = G = c = 1$.

2. Model, scalar equation of motion and numerical method

We consider the EsGB gravity with a positive cosmological constant of which the action is

$$S = \frac{1}{16\pi} \int d^4x \sqrt{-g} [R - 2\Lambda - 2\nabla_\mu \phi \nabla^\mu \phi - m^2 \phi^2 - f(\phi) \mathcal{L}_{\text{GB}}], \quad (1)$$

where R is the Ricci scalar, ϕ is the scalar field with mass m , $f(\phi)$ is the coupling function and $\mathcal{L}_{\text{GB}} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ is the GB term.¹ It is noticed that the properties of

¹ The Gauss–Bonnet term is the simplest higher curvature correction to Einstein's theory. The GB terms are related to the (in)stability issues of black holes, and physicists used to know that the GB terms have no contribution to the dynamic of the system when the dimension is less than five, however, recently the GB terms in four-dimensional spacetime were reconsidered. The general properties of black holes in Einstein–Gauss–Bonnet gravity were studied in [48–57] and references therein.

the EsGB theory closely depend on the forms of the coupling function $f(\phi)$. In order to admit the SdS black hole as a background solution, $f(\phi)$ should fulfill the conditions $\frac{df(\phi)}{d\phi}|_{\phi=0} = 0$ and $\frac{d^2f(\phi)}{d\phi^2}|_{\phi=0} = b > 0$ [7]. Additionally, one could consider the scalar field to be vanishing at infinity distance and normalize the parameter b to be unity. The simplest form of the coupling function which satisfies the aforementioned conditions is $f(\phi) = a_0 - \alpha\phi^2$.

A trivial solution of the scalar field is $\phi = 0$, then the scalar-free solution is described by the SdS black hole

$$ds^2 = -g(r)dt^2 + \frac{dr^2}{g(r)} + r^2(\sin^2\theta d\theta^2 + d\phi^2) \quad (2)$$

$$\text{with } g(r) = 1 - \frac{2M}{r} - \frac{\Lambda r^2}{3}, \quad (3)$$

where the constant M is the black hole mass. Subsequently, the GB term takes the form $\mathcal{L}_{\text{GB}} = \frac{48M^2}{r^6} + \frac{8\Lambda^2}{3}$. Solving $g(r) = 0$ could give us the event horizon (r_e) and cosmological horizon (r_c) of the SdS black hole

$$r_e = \frac{2}{\sqrt{\Lambda}} \cos \left[\frac{1}{3} \cos^{-1}(3M\sqrt{\Lambda}) + \frac{\pi}{3} \right],$$

$$r_c = \frac{2}{\sqrt{\Lambda}} \cos \left[\frac{1}{3} \cos^{-1}(3M\sqrt{\Lambda}) - \frac{\pi}{3} \right], \quad (4)$$

which are positive and real. It is noted that the above Killing horizons only hold when $3M\sqrt{\Lambda} < 1$, which is known as the Nariai limit. Beyond this limit, no black hole horizon exists; and when $3M\sqrt{\Lambda} = 1$, r_e and r_c are coincident.

Besides r_e and r_c , a third root to $g(r) = 0$ is denoted as r_0 which is negative and real. Using the three roots, $g(r)$ can be rewritten as $g(r) = \frac{\Lambda}{3r}(r - r_e)(r_c - r)(r - r_0)$. Then, the tortoise coordinate $r_* = \int g^{-1}(r) dr$ can be analytically evaluated as

$$r_*(r) = \frac{1}{2\kappa_e} \ln \left[\frac{r}{r_e} - 1 \right] - \frac{1}{2\kappa_c} \ln \left[1 - \frac{r}{r_c} \right] + \frac{1}{2\kappa_0} \ln \left[\frac{r}{r_0} - 1 \right], \quad (5)$$

where $\kappa_i = \frac{1}{2}|g'(r)|_{r=r_i}$ ($i = e, c, o$) is the surface gravity associated with each root.

Then, we consider a small scalar field perturbation on the background of the SdS black hole in the linear regime, which is governed by the covariant equation

$$\square\phi = \frac{\partial f(\phi)}{\partial\phi} \frac{\mathcal{L}_{\text{GB}}}{4} + \frac{m^2}{2}\phi \equiv \mu_{\text{eff}}^2\phi. \quad (6)$$

In equation (6), the box is the d'Alembertian operator and μ_{eff} is the effective mass.

$$\mu_{\text{eff}}^2 = \frac{m^2}{2} - \frac{\alpha}{2} \left(\frac{8\Lambda^2}{3} + \frac{48M^2}{r^6} \right). \quad (7)$$

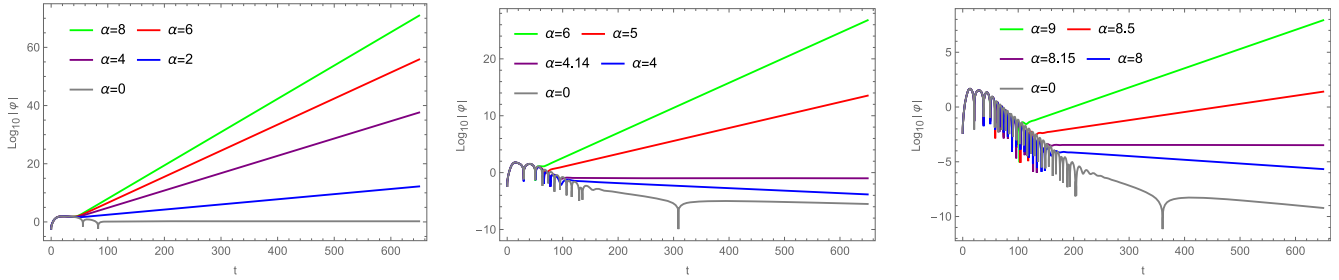


Figure 1. Time evolution of the massless scalar perturbation for $l = 0, 1, 2$ from left to right.

It is noticed that $\mu_{\text{eff}}^2 < 0$ requires $\alpha > 0$. This may trigger the tachyonic instability of the SdS black hole, which then leads to spontaneous scalarization in EsGB gravity [41].

To figure out the time evolution of the scalar field perturbation, we consider the scalar field as

$$\phi(t, r, \theta, \psi) = \sum_{lm} \frac{1}{r} R_{lm}(t, r) Y_{lm}(\theta, \psi), \quad (8)$$

where $Y_{lm}(\theta, \psi)$ denotes the scalar spherical harmonic function with angular momentum l and azimuthal number m . Then, we obtain that under the tortoise coordinate dr_* , each wave function R satisfies the equation

$$\left(-\frac{\partial^2}{\partial t^2} + \frac{\partial^2}{\partial r_*^2} - V_{\text{eff}}(r) \right) R(t, r) = 0, \quad (9)$$

where the effective potential has the form as

$$V_{\text{eff}}(r) = g(r) \left[\frac{l(l+1)}{r^2} + \frac{g'(r)}{r} + \mu_{\text{eff}}^2 \right], \quad (10)$$

where μ_{eff}^2 is defined in (7). The above analysis is not dependent of the azimuthal number m because of the spherical symmetry of the back hole.

Since it is not possible to find the analytical solution to these time-dependent wave equations, we have to solve it numerically. One simple efficient method is the discretization method applied in [58–60]. Then, by defining $R(r_*, t) = R(j\Delta r_*, i\Delta t) = R_{j,i}$, $V(r_*) = V(j\Delta r_*) = V_j$, one can write the wave equation (9) as

$$\begin{aligned} & -\frac{(R_{j,i+1} - 2R_{j,i} + R_{j,i-1}))}{\Delta t^2} \\ & + \frac{(R_{j+1,i} - 2R_{j,i} + R_{j-1,i}))}{\Delta r_*^2} \\ & - V_j R_{j,i} + O(\Delta t^2) + O(\Delta r_*^2) = 0. \end{aligned} \quad (11)$$

We choose the initial Gaussian distribution $R(r_*, t=0) = \exp\left[-\frac{(r_*-a)^2}{2b^2}\right]$ and $R(r_*, t<0) = 0$, then the evolution of R is controlled by

$$\begin{aligned} R_{j,i+1} = & -R_{j,i-1} + \frac{\Delta t^2}{\Delta r_*^2} R_{j+1,i} \\ & + R_{j-1,i} + \left(2 - 2\frac{\Delta t^2}{\Delta r_*^2} - \Delta t^2 V_j \right) R_{j,i}. \end{aligned} \quad (12)$$

In the numeric, we shall choose proper parameters $a = 10$ and $b = 3$ in the Gaussian profile for elegant precision. Additionally,

since the von Neumann stability conditions usually require that $\frac{\Delta t^2}{\Delta r_*^2} < 1$, so we fix $\frac{\Delta t^2}{\Delta r_*^2} = 0.5$ in the calculation.

For convenience, we shall set $M = 1$ and $\Lambda = 10^{-4}$, such that the event and cosmological horizons are located at $r_e \sim 2$ and $r_c \sim 172$, respectively. Also, we will focus on the behavior of the radial field $\varphi = R/r$ in the following.

3. The numerical results

We shall present the time evolution of the scalar perturbations affected by the GB coupling parameter, the angular momentum and the mass of the scalar field in the EsGB theory, and discuss the onset of the tachyonic instability from the dynamics.

3.1. The massless case

We first consider the massless scalar field with $m = 0$. The results of the evolutionary scalar perturbations are shown in figure 1 where we have fixed $l = 0, 1$ and $l = 2$ from left to right. In each plot, without the GB coupling, α , the scalar perturbation never grows up which means that the SdS black hole is a stable solution in the setup, as expected. New phenomena can be seen as the coupling exists, and the perturbation could grow up once α is larger than a critical value, α_c , implying the instability of the SdS black hole background. For $\alpha > \alpha_c$, the stronger coupling could make the instability occur earlier and also enhance the growth.

This influence of the GB coupling on the (in)stability of the scalar perturbation with different l can be understood from the behavior of effective potential. From (10), it is easy to retain that a larger α or smaller l will suppress the effective potential with a fixed radius near the event horizon. The profile of effective potential is explicitly shown in figure 2. In each case, the effective potential outside the event horizon is always positive when α is smaller than a certain value, α_* . However, as the coupling parameter increases to be larger than α_* , a negative potential well would form in the vicinity of the outer of the event horizon, and this well becomes deeper as α further increases. By carefully comparing the critical coupling parameter in figures 1 and 2, we find that $\alpha_* < \alpha_c$ for all cases. This is reasonable because a negative potential well is not a sufficient condition for occurrence of the instability. When the potential well is negative enough, the scalar field could gather near the event horizon and finally trigger the instability of the

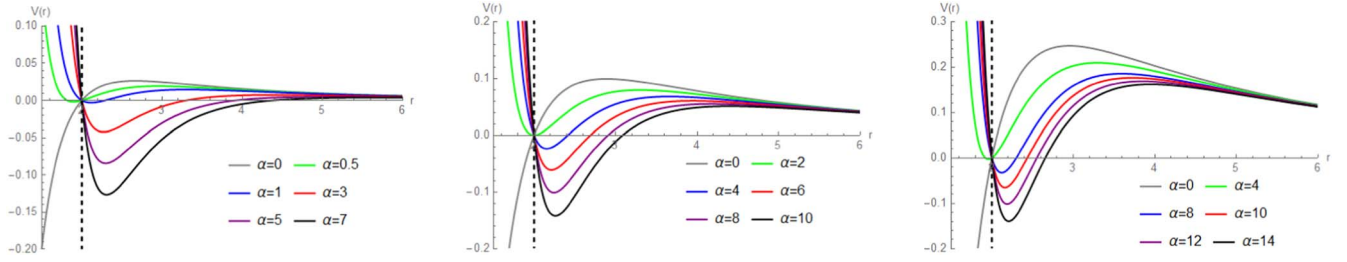


Figure 2. The profile of the effective potential as a function of radial coordinate for $l = 0, 1, 2$ from left to right. The vertical dashed line in each plot denotes the location of the event horizon.

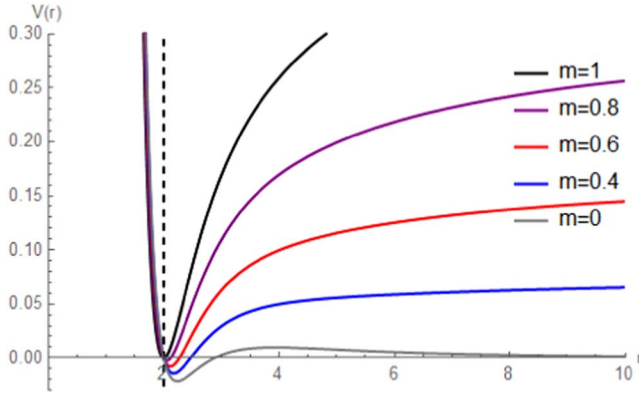


Figure 3. The effective potential as a function of the radial coordinate with different scalar mass for $l = 0$ and $\alpha = 2$. The vertical dashed line denotes the location of the event horizon.

Table 1. The critical coupling for different momentum angulars.

ℓ	0	1	2	3	4	5
α_c	—	4.14	8.16	13.50	20.20	28.32

background. By further studying the time evolution for more general modes of the perturbation, we figure out the critical GB coupling for the onset of the tachyonic instability, please see table 1. It is shown that the modes with larger l correspond to stronger critical GB coupling.

3.2. The massive case

We then turn on the mass of the scalar field, and study the onset of the tachyonic instability in $m - \alpha$ space. To this end, we focus on $l = 0$ mode without loss of generality.

According to the expressions (10) and (7), it is obvious that the larger mass could enhance the effective potential which is different from that of α . The effects of the selected mass on the potential are explicitly shown in figure 3. With fixed $\alpha = 2$, as the mass of the scalar field increases, the negative potential well becomes shallower and finally disappears. The related evolution of the scalar field is present in figure 4. It is clear that though the perturbation grows up and triggers instability for massless perturbation, when the mass is larger than a critical value, m_c , the perturbation would decay as the time passes, implying that the SdS background is stable under those perturbations.

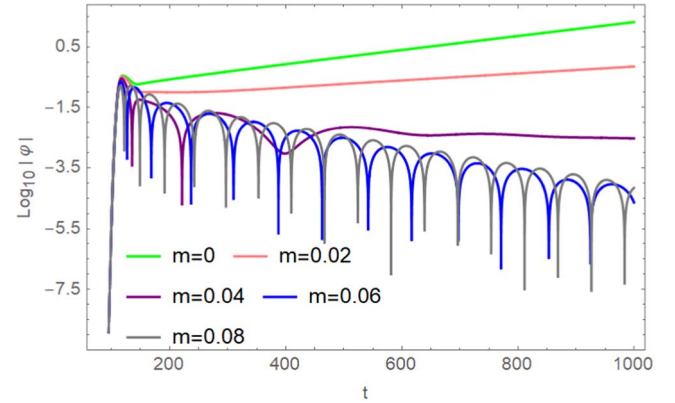


Figure 4. Time evolution of massive scalar field with $\alpha = 2$.

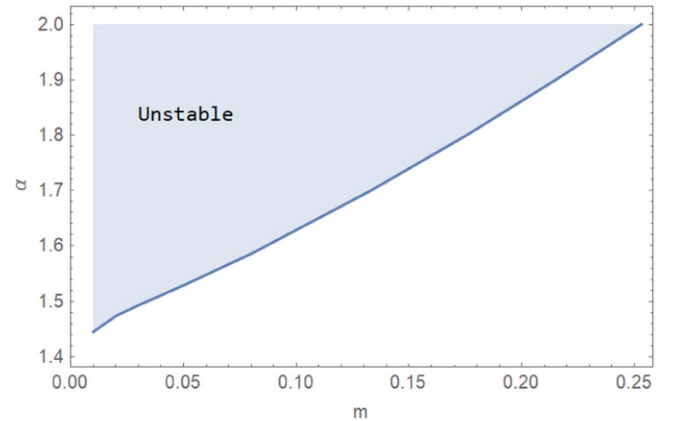


Figure 5. Stable/unstable region in $m - \alpha$ space for $\ell = 0$ mode.

We could find the corresponding m_c for each α . The complete picture of the onset of the tachyonic instability in $m - \alpha$ space is shown in figure 5, which clearly exhibits the aforementioned phenomena. It is noted that similar phenomena could be observed for modes with other angular momentum l .

4. Conclusions

In this paper, we studied the wave dynamics of the massive scalar field perturbation on a SdS black hole in a special EsGB theory. We fixed the black hole mass $M = 1$ and the cosmological constant $\Lambda = 10^{-4}$ and focused on the effect of the GB coupling and mass of the scalar field. When

$\alpha < \alpha_c$, the scalar field perturbation would decay as time passes, but the perturbation grows up when $\alpha > \alpha_c$. This growth in the scalar field perturbation may lead to the instability of the background black hole such that a scalarized hairy solution may form. So α_c indicates the onset of tachyonic instability. Moreover, the mass of the scalar field would suppress its growth, such that a larger mass could enhance the critical GB coupling strength, which agrees with the discussion in [44, 46]. The underlying reason is that a larger α corresponds to a more deeper negative potential well, which is the sufficient condition of the tachyonic instability. Meanwhile, a larger ℓ and m provide a positive contribution to the effective potential and enhance the negative potential well.

Note that here we studied the time evolution of the scalar field perturbation in the time domain with a small fixed cosmological constant. To have a complete understanding of the propagation, it is interesting to study the quasinormal frequencies as we mentioned in the introduction. It is known that only the discrete eigenfrequency, $\omega = \omega_R + i\omega_I$, satisfies the perturbation equation and related boundary conditions [61, 62]. Once $\omega_I > 0$, the perturbation will grow up, implying that the black hole is unstable under this perturbation. Moreover, the above analysis deserves to extend wider arrangements of the model parameters.

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References

- [1] Abbott B P *et al* (LIGO Scientific and Virgo) 2016 Observation of gravitational waves from a binary black hole merger *Phys. Rev. Lett.* **116** 061102
- [2] Abbott B P *et al* (LIGO Scientific and Virgo) 2016 GW151226: Observation of gravitational waves from a 22-solar-mass binary black hole coalescence *Phys. Rev. Lett.* **116** 241103
- [3] Akiyama K *et al* (Event Horizon Telescope) 2019 First M87 event horizon telescope results. I. The shadow of the supermassive black hole *Astrophys. J. Lett.* **875** L1
- [4] 2001 *String Theory* Polchinski J (ed) (Cambridge: Cambridge University Press) Vol. 1 & 2
- [5] Antoniou G, Bakopoulos A and Kanti P 2018 Evasion of no-hair theorems and novel black-hole solutions in Gauss-Bonnet theories *Phys. Rev. Lett.* **120** 131102
- [6] Silva H O, Sakstein J, Gualtieri L, Sotiriou T P and Berti E 2018 Spontaneous scalarization of black holes and compact stars from a Gauss-Bonnet coupling *Phys. Rev. Lett.* **120** 131104
- [7] Doneva D D and Yazadjiev S S 2018 New Gauss-Bonnet black holes with curvature-induced scalarization in extended scalar-tensor theories *Phys. Rev. Lett.* **120** 131103
- [8] Herdeiro C A R and Radu E 2019 Black hole scalarization from the breakdown of scale invariance *Phys. Rev. D* **99** 084039
- [9] Brihaye Y, Herdeiro C and Radu E 2019 The scalarised Schwarzschild-NUT spacetime *Phys. Lett. B* **788** 295–301
- [10] Minamitsuji M and Ikeda T 2019 Scalarized black holes in the presence of the coupling to Gauss-Bonnet gravity *Phys. Rev. D* **99** 044017
- [11] Silva H O, Macedo C F B, Sotiriou T P, Gualtieri L, Sakstein J and Berti E 2019 Stability of scalarized black hole solutions in scalar-Gauss-Bonnet gravity *Phys. Rev. D* **99** 064011
- [12] Andreou N, Franchini N, Ventagli G and Sotiriou T P 2019 Spontaneous scalarization in generalised scalar-tensor theory *Phys. Rev. D* **99** 124022
- Andreou N, Franchini N, Ventagli G and Sotiriou T P 2020 Erratum: Spontaneous scalarization in generalized scalar-tensor theory [Phys. Rev. D 99, 124022 (2019)] *Phys. Rev. D* **101** 109903
- [13] Minamitsuji M and Ikeda T 2019 Spontaneous scalarization of black holes in the Horndeski theory *Phys. Rev. D* **99** 104069
- [14] Peng Y 2020 Spontaneous scalarization of Gauss-Bonnet black holes surrounded by massive scalar fields *Phys. Lett. B* **807** 135569
- [15] Liu H S, Lu H, Tang Z Y and Wang B 2021 Black hole scalarization in Gauss-Bonnet extended Starobinsky gravity *Phys. Rev. D* **103** 084043
- [16] Doneva D D, Staykov K V, Yazadjiev S S and Zheleva R Z 2020 Multiscalar Gauss-Bonnet gravity: Hairy black holes and scalarization *Phys. Rev. D* **102** 064042
- [17] Astefanesei D, Herdeiro C, Oliveira J and Radu E 2020 Higher dimensional black hole scalarization *J. High Energy Phys. JHEP09(2020)186*
- [18] Cañate P and Perez Bergliaffa S E 2020 Novel exact magnetic black hole solution in four-dimensional extended scalar-tensor-Gauss-Bonnet theory *Phys. Rev. D* **102** 104038
- [19] Hunter C L and Smith D J 2022 Novel Hairy Black Hole Solutions in Einstein-Maxwell-Gauss-Bonnet-Scalar Theory *Int. J. Mod. Phys. A* **37** (9) 2250045
- [20] Bakopoulos A, Kanti P and Pappas N 2020 Existence of solutions with a horizon in pure scalar-Gauss-Bonnet theories *Phys. Rev. D* **101** 044026
- [21] Bakopoulos A, Kanti P and Pappas N 2020 Large and ultracompact Gauss-Bonnet black holes with a self-interacting scalar field *Phys. Rev. D* **101** 084059
- [22] Lin K, Zhang S, Zhang C, Zhao X, Wang B and Wang A 2020 No static regular black holes in Einstein-complex-scalar-Gauss-Bonnet gravity *Phys. Rev. D* **102** 024034
- [23] Brihaye Y, Hartmann B, Aprile N P and Urrestilla J 2020 Scalarization of asymptotically anti-de Sitter black holes with applications to holographic phase transitions *Phys. Rev. D* **101** 124016
- [24] Guo H, Kiorpelidi S, Kuang X M, Papantonopoulos E, Wang B and Wu J P 2020 Spontaneous holographic scalarization of black holes in Einstein-scalar-Gauss-Bonnet theories *Phys. Rev. D* **102** 084029
- [25] Tang Z Y, Wang B, Karakasis T and Papantonopoulos E 2021 Curvature scalarization of black holes in $f(R)$ gravity *Phys. Rev. D* **104** 064017
- [26] Collodel L G, Kleihaus B, Kunz J and Berti E 2020 Spinning and excited black holes in Einstein-scalar-Gauss-Bonnet theory *Class. Quant. Grav.* **37** 075018
- [27] Dima A, Barausse E, Franchini N and Sotiriou T P 2020 Spin-induced black hole spontaneous scalarization *Phys. Rev. Lett.* **125** 231101
- [28] Herdeiro C A R, Radu E, Silva H O, Sotiriou T P and Yunes N 2021 Spin-induced scalarized black holes *Phys. Rev. Lett.* **126** 011103
- [29] Berti E, Collodel L G, Kleihaus B and Kunz J 2021 Spin-induced black-hole scalarization in Einstein-scalar-Gauss-Bonnet theory *Phys. Rev. Lett.* **126** 011104

- [30] Bao Y, Guo H and Kuang X M 2021 Excited states of holographic superconductor with scalar field coupled to Gauss-Bonnet invariance *Phys. Lett. B* **822** 136646
- [31] Guo H, Kuang X M, Papantonopoulos E and Wang B 2021 Horizon curvature and spacetime structure influences on black hole scalarization *Eur. Phys. J. C* **81** 842
- [32] Yang Z H, Fu G, Kuang X M and Wu J P 2022 Instability of de-Sitter black hole with massive scalar field coupled to Gauss-Bonnet invariant and the scalarized black holes *Eur. Phys. J. C* **82** 868
- [33] Zhang S J, Wang B, Papantonopoulos E and Wang A 2023 Magnetic-induced spontaneous scalarization in dynamical Chern-Simons gravity *Eur. Phys. J. C* **83** 97
- [34] Liu Y, Zhang C Y, Chen Q, Cao Z, Tian Y and Wang B 2023 Critical scalarization and descalarization of black holes in a generalized scalar-tensor theory *Sci. China Phys. Mech. Astron.* **66** 100412
- [35] Myung Y S 2023 Two instabilities of Schwarzschild-AdS black holes in Einstein-Weyl-scalar theory *Eur. Phys. J. C* **83** 902
- [36] Perlmutter S et al (Supernova Cosmology Project) 1999 Measurements of Ω and Λ from 42 high redshift supernovae *Astrophys. J.* **517** 565–86
- [37] Riess A G et al (Supernova Search Team) 1998 Observational evidence from supernovae for an accelerating universe and a cosmological constant *Astron. J.* **116** 1009–38
- [38] Strominger A 2001 The dS/CFT correspondence *J. High Energy Phys.* **JHEP10(2021)034**
- [39] Witten E 2001 Quantum gravity in de Sitter space arXiv:[hep-th/0106109](https://arxiv.org/abs/hep-th/0106109)
- [40] Bakopoulos A, Antoniou G and Kanti P 2019 Novel black-hole solutions in Einstein-Scalar-Gauss-Bonnet theories with a cosmological constant *Phys. Rev. D* **99** 064003
- [41] Brihaye Y, Herdeiro C and Radu E 2020 Black hole spontaneous scalarisation with a positive cosmological constant *Phys. Lett. B* **802** 135269
- [42] Castineiras J, Crispino L C B and Filho D P M 2007 Source coupled to the massive scalar field orbiting a stellar object *Phys. Rev. D* **75** 024012
- [43] Maselli A, Franchini N, Gualtieri L, Sotiriou T P, Barsanti S and Pani P 2022 Detecting fundamental fields with LISA observations of gravitational waves from extreme mass-ratio inspirals *Nat. Astron.* **6**
- [44] Doneva D D, Staykov K V and Yazadjiev S S 2019 Gauss-Bonnet black holes with a massive scalar field *Phys. Rev. D* **99** 104045
- [45] Macedo C F B, Sakstein J, Berti E, Gualtieri L, Silva H O and Sotiriou T P 2019 Self-interactions and spontaneous black hole scalarization *Phys. Rev. D* **99** 104041
- [46] Doneva D D, Collodel L G, Krüger C J and Yazadjiev S S 2020 Spin-induced scalarization of Kerr black holes with a massive scalar field *Eur. Phys. J. C* **80** 1205
- [47] Vishveshwara C V 1970 Scattering of Gravitational Radiation by a Schwarzschild Black-hole *Nature* **227** 936–8
- [48] Boulware D G and Deser S 1985 String generated gravity models *Phys. Rev. Lett.* **55** 2656
- [49] Wiltshire D L 1988 Black holes in string generated gravity models *Phys. Rev. D* **38** 2445
- [50] Cai R G 2002 Gauss-Bonnet black holes in AdS spaces *Phys. Rev. D* **65** 084014
- [51] Neupane I P 2004 Thermodynamic and gravitational instability on hyperbolic spaces *Phys. Rev. D* **69** 084011
- [52] Ge X H and Sin S J 2009 Shear viscosity, instability and the upper bound of the Gauss-Bonnet coupling constant *J. High Energy Phys.* **JHEP05(2009)051**
- [53] Kuang X M, Wang B and Wu J P 2012 Dipole coupling effect of holographic fermion in the background of charged Gauss-Bonnet AdS black hole *J. High Energy Phys.* **JHEP07(2012)125**
- [54] Aránguiz L, Kuang X M and Miskovic O 2016 Topological black holes in pure Gauss-Bonnet gravity and phase transitions *Phys. Rev. D* **93** 064039
- [55] Kuang X M, Saavedra J and Övgün A 2017 The effect of the Gauss-Bonnet term to Hawking Radiation from arbitrary dimensional Black Brane *Eur. Phys. J. C* **77** 613
- [56] Glavan D and Lin C 2020 Einstein-Gauss-Bonnet gravity in four-dimensional spacetime *Phys. Rev. Lett.* **124** 081301
- [57] Ge X H and Sin S J 2020 Causality of black holes in 4-dimensional Einstein-Gauss-Bonnet-Maxwell theory *Eur. Phys. J. C* **80** 695
- [58] Zhu Z, Zhang S J, Pellicer C E, Wang B and Abdalla E 2014 Stability of Reissner-Nordström black hole in de Sitter background under charged scalar perturbation *Phys. Rev. D* **90** 044042
- [59] Zhang S J, Wang B, Wang A and Saavedra J F 2020 Object picture of scalar field perturbation on Kerr black hole in scalar-Einstein-Gauss-Bonnet theory *Phys. Rev. D* **102** 124056
- [60] Zhang S J 2021 Massive scalar field perturbation on Kerr black holes in dynamical Chern-Simons gravity *Eur. Phys. J. C* **81** 441
- [61] Berti E, Cardoso V and Starinets A O 2009 Quasinormal modes of black holes and black branes *Class. Quant. Grav.* **26** 163001
- [62] Konoplya R A and Zhidenko A 2011 Quasinormal modes of black holes: From astrophysics to string theory *Rev. Mod. Phys.* **83** 793–836